Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_budget_matched_no_replay.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	41.85882	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	75.76836	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1366.45918	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	205.2921	1	0	0	0.959254	14.3
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	1.0	0.070834	0.070834	207.65304	2	0	0	0.960795	13.9
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	142.31612	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 41.85882.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 75.76836.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1366.45918.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 205.2921.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.067412.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070834.
- Held-out runtime ms: 207.65304.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.004209.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.098364, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.095616, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.058186, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.056154, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.045849, errors=[]

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 142.31612.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Condition	Feasibility Rate	Heldout Feasible Gap	Runtime (ms)	Robustness Dispersion	Learned (Novelty & Complexity)
Baseline Nearest Neighbor Constructive	100%	0.2734	41.86	0.0028	None (novelty = 0, complexity = 0)
Baseline Clarke-Wright Savings	100%	0.0738	75.77	0.0078	None (novelty = 0, complexity = 0)
Baseline Regret Insertion Local Search	100%	0.4664	1366.46	0.0869	None (novelty = 0, complexity = 0)
Direct Generate + One Repair	0%	0	205.29	0	Learned (high novelty 0.96, complexity 14.3) but infeasible
Budget-Matched Independent Search (No Replay)	100%	**0.0708 (best)**	207.65	0.0042	Learned (high novelty 0.96, complexity 13.9)
Replay Solver Evolution	100%	0.2734	142.32	0.0028	None (novelty = 0, complexity = 0)

Key Insights

- **Feasibility:** All methods except *Direct Generate + One Repair* achieved 100% feasibility on held-out data.
- **Objective Gap:** The *budget-matched no replay* condition attained the lowest heldout feasible gap (0.0708), slightly better than the Clarke-Wright baseline (0.0738). Replay condition matched nearest neighbor gap but was worse than budget-matched no replay.
- **Runtime:** Nearest neighbor was fastest, but with high gap. Budget-matched and replay methods had moderate runtimes (~140-208ms). Regret insertion was very slow (~1366ms).
- **Robustness:** Budget-matched no replay had low robustness dispersion (0.0042), near baseline levels.
- **Learning / Novelty:** Only budget-matched no replay and direct generate + repair exhibited high code novelty and complexity. Direct generate failed feasibility.
- **Replay vs No Replay:** Replay did not outperform the budget-matched no replay control for gap or robustness; budget-matched no replay was clearly better.
- **Learned Conditions vs Baselines:** No learned condition, except budget-matched no replay, outperformed fixed baselines. Direct generate approach was infeasible.

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Conclusion

- The **budget-matched independent search without replay** condition clearly **outperformed fixed baselines** in gap while maintaining feasibility, robustness, and reasonable runtime.
- The **replay-aware iterative search did not surpass** the budget-matched no-replay baseline.
- Direct synthesis with repair failed feasibility and thus did not improve over baselines.
- Overall, budget-matched independent search is the preferred learned approach in phase-9 closeout for CVRP.